

RHEL + Podman standards v0.1

RHEL + Podman standards v0.1 (banking baseline)

SOW item 2 — RHEL standards & best practices. **Prescriptive defaults** for this client. Client platform approves and implements. Contract: ; discovery alignment: . **Broader platform standards (access, DR, naming, gates):** . **Industry authorities (CIS, NIST, FFIEC, Azure, Red Hat):** .

1. Architecture

Rule	Standard
VMs per env	Two: GitHub runner + Podman runtime
Golden image	Layer 1 SIG — OS only; no app workloads
Deploy	Layer 2 cloud-init/Ansible; Layer 3 GitHub Actions
Container runtime	Podman only — no Docker
Registry	ACR Premium (<code>acr-baseline</code> module) — UAMI <code>AcrPush</code> / <code>AcrPull</code> , private endpoint, quarantine
Image build	Runner only — <code>build/images.manifest.json</code> + <code>build-push-images.sh</code> → ACR

2. Identity (IAM)

Rule	Standard
Human access	Entra ID SSH or per-user accounts — no shared admin
Developer az CLI	PIM eligible roles on dev sub only → <code>scripts/az-developer-login.sh</code> ; see
VM auth	One UAMI per VM — never shared across runner/runtime
Runner UAMI	<code>AcrPush</code> , <code>Key Vault Secrets User</code>
Runtime UAMI	<code>AcrPull</code> , <code>Key Vault Secrets User</code> only
Secrets	Key Vault — no secrets in git, env files on disk, or golden image
Azure CLI on UAT/prod VM	<code>az login --identity -u <uami-id></code> only
Azure CLI on dev (human)	PIM activate → <code>az-login-developer.sh</code> — not UAMI

3. Network (NSGs)

Rule	Standard
NSG per VM role	<code>nsg-baseline</code> module — deny all inbound default
SSH inbound	Bastion subnet CIDR only
Runtime inbound	APIM subnet (or internal consumers) only
Public IP	Denied (Azure Policy sample provided)
Private IP	Static — document in firewall allow-list
Egress	Hub firewall allow-list — see

4. Storage

Rule	Standard
Runtime data	Premium managed data disk — UUID in <code>fstab</code> via <code>mount-data-disk.sh</code>
Podman graph	On data disk (<code>/var/lib/containers</code>)
Shared files	Azure Files NFS optional — UAMI RBAC, private endpoint
Encryption	SSE default; CMK if policy requires

5. OS hardening (Layer 1)

OS baseline (single primary standard)

Rule	Standard
Primary baseline	CIS RHEL 9 Level 1 — OpenSCAP remediate + verify in AIB; evidence in <code>/opt/compliance/artifacts/</code>
DISA STIG	Not default. Use only if GRC mandates — assess with OpenSCAP STIG profile in non-prod ; expect Podman/app compatibility testing and exception process
Do not stack	CIS L1 + STIG on same image without GRC approval — overlapping controls, higher break risk, duplicate remediation
Compliance mapping	Map CIS controls to FFIEC / NIST 800-53 via — consultant informational only; GRC attests

GRC decision (client records): ☐ CIS L1 sufficient ☐ STIG required for these hosts

Technical controls

Control	Implementation
Trusted Launch	Secure Boot + vTPM
SELinux	Enforcing
SSH	No root, no passwords
auditd	Enabled — identity, sudo, execve, Podman paths; audisp → rsyslog
journald / rsyslog	Capped persistent journal; imjournal rate limits; daily logrotate + <code>/var/log/archive</code> retention
Log archiving	90-day rotated logs; 365-day archive; audit-forward log for SIEM path
Patching	dnf-automatic (<code>security</code> only, <code>reboot</code> never); monthly <code>dnf update --security</code> window for validation

Vulnerability assessment gates (OS / VM layer)

Gate	Rule	Owner
Dev	Remediate P0 findings from pen test / Arctic Wolf / continuous scan on Linux before UAT template or SIG promotion	Client security
UAT	No UAT Linux build from golden image while open P0 on OS hardening, SSH, or shared-admin issues	Client platform
Prod	Re-scan or security sign-off before first prod Linux cutover	Client security (Client security)
Consultant	Advise prioritization and map findings to this standards doc — does not execute pen test or remediate	Consultant

6. CI/CD (Layer 3)

Rule	Standard
CI platform	GitHub Actions self-hosted runner
Build	On runner VM only
Runtime	Pull-only — no podman build on UAT/prod
Ingress	nginx reverse-proxy sidecar — sole published port; app containers internal only — decision:
Workload units	Quadlet (<code>.container</code> , <code>.network</code>) under <code>/etc/containers/systemd/</code>
Reference stack	Gorobi, ingestion, FastAPI APIs, Dagster + nginx —
Promote	GitHub Environments with required reviewers for prod
Reference workflow	<code>.github/workflows/build-deploy.yml</code> , <code>build-images.yml</code>
Build layer doc	

7. Monitoring & security

Item	Owner	Implementation
Azure Monitor	Client platform (SIEM path)	Layer 1: rsyslog + AMA prereqs in golden image (azure_monitor.yml). Layer 2: monitor-baseline Terraform module — DCR (syslog auth/daemon/kern + perf counters) → Log Analytics, AzureMonitorLinuxAgent extension, UAMI Monitoring Metrics Publisher on DCR
Logic Monitor	Client platform	Separate from Azure Monitor — layer-2 bootstrap script on image (enable-logicmonitor-collector.sh)
EDR	Client SOC	Install via approved Ansible when vendor ready
Pen test / vuln scan	Client security	Arctic Wolf, annual pen test, or equivalent — see §5 gates; consultant advises priority only
STIG (if mandated)	Client GRC + platform	Separate assessment track — not combined with CIS L1 without approval

Coexistence: Azure Monitor feeds the bank Log Analytics / Sentinel path; Logic Monitor remains the Client platform-owned operational dashboard. Both agents may run on the same VM.

Layer 1 golden image does not store: workspace keys, DCR IDs, or AMA association — those are Layer 2 Terraform only.

Industry references

Full authority map:

Section	Primary sources
§1 Architecture	Azure CAF Landing Zone NIST SP 800-190
§2 Identity	Entra managed identities Entra SSH for Linux PIM
§3 Network	NSG overview MCSB network controls
§4 Storage	Azure managed disks Azure Files NFS
§5 OS hardening	CIS RHEL 9 Benchmark OpenSCAP RHEL 9 Security Hardening DISA STIGs
§6 CI/CD	Podman Quadlet GitHub self-hosted runners GitHub Environments
§7 Monitoring	Azure Monitor Agent DCR overview